2. Scope of Work

2.1. Background

Pimpri Chinchwad Municipal Corporation (PCMC) increasingly is adopting digital platforms to enhance citizen service delivery, improve transparency, and reduce dependency on physical service centers. With widespread adoption of smartphones and WhatsApp across all demographic groups, there is a significant opportunity to leverage this platform as a primary channel for municipal service delivery. Pimpri Chinchwad Municipal Corporation (PCMC) aims to deploy a WhatsApp-based AI chatbot, "Sarathi AI," to provide citizens with round-the-clock access to municipal information and services. The initiative is designed to simplify interactions between citizens and the municipal administration, reduce footfall at ward offices, improve grievance redressal efficiency, and strengthen trust in civic governance through a secure, multilingual, and user-friendly digital interface.

Pimpri Chinchwad Municipal Corporation (PCMC) intends to design, develop, implement, integrate, operate, maintain, and continuously enhance a comprehensive AI-powered citizen engagement and service delivery platform named Sarathi AI, accessible through WhatsApp and other approved digital channels.

The selected bidder shall provide an end-to-end turnkey solution including application development, AI engine, WhatsApp integration, multilingual services, citizen survey capabilities, analytics, dashboards, integrations, security, training, support and operations throughout the contract period. The solution shall provide a single digital interface for citizen interaction with PCMC departments and services.

2.2. Objective

The primary objective of the Sarathi AI WhatsApp Chatbot is to provide 24×7, simple, multilingual, and mobile-first access to municipal services for citizens of PCMC. The solution aims to improve ease of living by enabling citizens to obtain service information, track applications, lodge and monitor grievances, receive timely notifications, and access payment-related information directly through WhatsApp. The initiative seeks to enhance transparency, accountability, and responsiveness in service delivery, reduce physical visits and manual interventions, encourage digital payments and feedback, and enable data-driven decision-making for the municipal administration while ensuring compliance with applicable data privacy and security regulations.

The objectives of the project are to:

- Provide 24x7 citizen services through WhatsApp.
- Improve accessibility and citizen convenience.
- Enhance grievance redressal efficiency.
- Improve transparency and accountability.
- Increase digital adoption and self-service.
- Enable multilingual citizen interaction.
- Support AI-driven service discovery and assistance.
- Facilitate citizen feedback and survey initiatives.
- Enable data-driven governance through analytics and dashboards.

- Reduce physical visits to municipal offices.
- Improve operational efficiency across municipal departments.

2.3. Broad Scope of the Selected Bidder

The selected bidder shall be responsible for:

- Requirement gathering and business process study.
- Solution architecture and detailed design.
- Application development and customization.
- WhatsApp Business Platform integration.
- AI and LLM integration.
- Knowledge Base creation and management.
- Integration with PCMC applications.
- Deployment and go-live.
- Security and compliance implementation.
- User acceptance testing.
- Training and knowledge transfer.
- Operations and maintenance.
- Continuous optimization of AI responses.
- Support for future onboarding of services and departments.

The engagement shall be on a turnkey basis and include all software, licenses, subscriptions, APIs, AI consumption, Tokens, WhatsApp services, third-party services and recurring costs unless specifically designated as PCMC responsibility.

2.4. Detailed Scope of Work

The scope of work includes the design, development, integration, deployment, and support of a comprehensive WhatsApp-based AI chatbot for PCMC. The solution shall provide end-to-end citizen services including service information and guided assistance, application and certificate status tracking, grievance registration and real-time tracking, payment enablement, and proactive notifications related to utilities, public safety, and civic alerts. The chatbot shall support multilingual interaction and simple, menu-driven navigation to ensure accessibility across all user groups. The scope further includes citizen engagement tools such as feedback mechanisms, surveys and awareness campaigns, along with backend administrative features including integration with existing complaint management systems, ERP and payment systems, dashboards and analytics, SLA monitoring, escalation mechanisms, and compliance with data protection requirements. Future-ready capabilities such as AI-based intent detection and GIS-based complaint mapping may also be implemented to enhance intelligence, planning, and efficiency.

1.2 Functional Scope

1.2.1 Application Design, Development and Deployment

The bidder shall design, develop, test, deploy and commission the Sarathi AI platform including:

A. Citizen Interfaces

- WhatsApp based citizen interface.
- Responsive web interface.
- Administrative portal.
- Operator support interface.
- Department dashboards.

B. Core Functional Components

- Citizen service workflows.
- Grievance management workflows.
- Payment workflows.
- Certificate and application tracking.
- Survey and feedback workflows.
- Broadcast communication workflows.
- Audit and monitoring framework.

C. Technical Deliverables

- Application architecture.
- Database design.
- API framework.
- Security architecture.
- Integration architecture.
- Deployment architecture.
- Disaster recovery design.

1.2.2 AI, NLP and Orchestration Platform

The bidder shall establish a comprehensive AI ecosystem comprising:

A. AI Components

- AI chatbot engine.
- NLP engine.
- Machine learning services.
- Prompt orchestration layer.
- Agent framework.
- MCP server architecture.
- Knowledge base engine.
- Retrieval Augmented Generation (RAG) framework.
- Vector database.

B. AI Functionalities

- Intent detection.
- Entity extraction.
- Context-aware conversations.

- FAQ automation.
- Service discovery.
- Response generation.
- Sentiment analysis.
- Content moderation.
- Duplicate complaint identification.
- AI-generated summaries.
- Recommendation engine.
- Predictive insights.

The solution shall support OpenAI, Azure OpenAI, Gemini or equivalent enterprise-grade models.

1.2.3 WhatsApp Business Platform

The bidder shall provide complete WhatsApp service enablement including:

A. WhatsApp Infrastructure

- Official WhatsApp Business API integration.
- BSP services.
- WhatsApp onboarding.
- Verification support.
- Number provisioning support.
- Webhook configuration.

B. Messaging Services

- Authentication messages.
- Utility messages.
- Service messages.
- Session messages.
- Broadcast messages.
- Awareness campaigns.
- Emergency notifications.
- Citizen engagement campaigns.

C. Citizen Journeys

- Menu-based navigation.
- Interactive buttons.
- Forms and workflows.
- Human handoff.
- Live operator routing.
- Multilingual conversations.

All WhatsApp conversation and recurring platform costs shall be included in the bidder's scope.

1.2.4 Citizen Services Module

The chatbot shall enable citizens to:

A. Information Services

- Access information regarding municipal services.
- Access information and locations regarding facilities (e.g. Gardens, Schools, BRT, Hospitals, etc.)
- View eligibility criteria for the schemes/programs.
- RTS and RTI Services
- Understand required documents.
- View timelines and process steps.

B. Application Tracking

- RTS Services
- Birth certificate status.
- Death certificate status.
- Marriage certificate status.
- Property-related applications.
- Trade licenses.
- Hoarding permissions.
- Hawker licenses.
- Any future PCMC services.

C. Office Information

- Ward offices.
- Officer contact details.
- Working hours.
- Department information.

1.2.5 Grievance Management System

The solution shall support:

A. Complaint Registration

- Text submission.
- Voice submission.
- Image upload.
- Video upload.
- Geo-location sharing.

B. Automated Processing

- Complaint categorization.
- Complaint prioritization.
- Department mapping.
- Workflow routing.

- Escalation generation.

C. Complaint Lifecycle

- Acknowledgement.
- Status tracking.
- Action updates.
- Closure communication.
- Feedback collection.

1.2.6 Payment Enablement Services

The platform shall support:

- Property tax information.
- Water bill information.
- Municipal fee information.
- Demand notice retrieval.
- Secure payment links.
- Payment gateway integration.
- Payment status updates.
- Digital receipts.
- Transaction history.
- Reconciliation support.

1.2.7 Citizen Survey and Feedback Management

The bidder shall implement a dedicated survey module.

A. Survey Functions

- Survey creation.
- Questionnaire management.
- Multi-language surveys.
- Citizen segmentation.
- Campaign scheduling.
- Ward-wise surveys.
- Satisfaction surveys.
- Polls and consultations.

B. Survey Types

- Service satisfaction surveys.
- Post-grievance surveys.
- Citizen opinion polls.
- Governance feedback studies.
- Public consultation campaigns.

C. Analytics

- Satisfaction scores.
- NPS analysis.
- Response analytics.
- Sentiment analysis.
- Department-wise insights.
- Ward-wise analytics.
- Trend analysis.

1.2.8 Knowledge Base and Self-Service Platform

The bidder shall develop and maintain:

- FAQ repository.
- Service knowledge base.
- Policy repository.
- Circular repository.
- Citizen help content.
- AI searchable knowledge repository.

The knowledge base shall support continuous updates throughout the contract period.

Knowledge Base Ownership – Details to be specified:

- Prompt library ownership
- Embeddings ownership
- Vector DB ownership
- Training datasets ownership

1.2.9 Multilingual and Voice Services

The platform shall support:

A. Languages

- Marathi.
- English.
- Additional Indian languages as required.

B. Voice Services

- Speech-to-text.
- Text-to-speech.
- Voice note understanding.
- Audio interaction support.

C. Translation

- Automatic language detection.
- Real-time translations.
- Multilingual AI responses.

1.2.10 OCR, Document and Image AI

The solution shall support:

- OCR services.
- Document reading.
- Image understanding.
- Citizen document uploads.
- Application document verification support.
- AI-powered extraction of information from uploaded files.

1.2.11 AI Governance

Bidders shall provide:

- Hallucination control
- Human escalation
- Prompt logging
- AI response audit trail
- Content moderation
- Safety controls

1.2.12 GIS and Geo-Spatial Intelligence

The bidder shall provide:

- Location capture.
- Geo-tagged complaints.
- Ward mapping.
- Zone mapping.
- Heat maps.
- Asset mapping.
- GIS analytics.
- Location-based routing

1.2.13 Integration Requirements

The bidder shall integrate with existing and future PCMC systems. Including but not limited to:

- GIS enabled ERP.
- Core and Non-Core Applications
- Grievance Management System.
- Property Tax System.
- Water Billing System.
- Payment Gateway.
- Document Repository.

- GIS Platform.
- e-Governance Applications.
- ICCC Systems.
- Citizen Databases.
- Third-party applications approved by PCMC.

All required APIs, middleware and connectors shall be included in bidder scope.

1.2.14 Dashboard, Analytics and Administration Portal

The solution shall provide:

A. Dashboards

- Citizen interaction dashboard.
- Service delivery dashboard.
- Survey dashboard.
- Grievance dashboard.
- Payment dashboard.
- AI usage dashboard.

B. Analytics

- Ward-wise analytics.
- Department-wise analytics.
- SLA monitoring.
- Citizen satisfaction score.
- Service consumption trends.
- Operational KPIs.

C. Administration

- User management.
- Role-based access control.
- Workflow configuration.
- Template management.
- Campaign management.

1.2.15 Security, Compliance and Audit

The bidder shall ensure compliance with:

- DPDP Act.
- Information Technology Act.
- Government cybersecurity guidelines.
- CERT-In requirements.
- PCMC security policies.

The scope shall include:

VAPT.

- Secure coding.
- Audit logs.
- Encryption.
- Access controls.
- Security monitoring.
- Compliance reporting

1.2.16 Training and Knowledge Transfer

The bidder shall provide:

- Administrator training.
- Department user training.
- Operator training.
- Technical training.
- Documentation.
- SOPs.
- User manuals.
- Configuration manuals.
- Technical handover documents